Teacher Guide — Newton's Laws of Motion

Physics | Grade 9-10

Period 1

Entry Ticket

Name one situation where something kept moving after you stopped pushing it.

Expected response: Any example of continued motion, e.g. a rolling ball.

Teacher Script

0-5 min: Entry ticket

Collect two or three answers without correcting them yet.

Board: Write the three best student examples in a column.

5-20 min: Inertia and the First Law

Introduce the idea using the student examples already on the board, then state the law formally.

Board: Write the formal statement beneath the examples.

- Anticipate: Why does a rolling ball eventually stop?

20-32 min: Activity

Run the activity; circulate and listen for the misconception.

32-37 min: Checkpoint

Pose the checkpoint question; take a show of hands before answers.

37-40 min: Exit ticket

Collect exit tickets at the door.

Blackboard Notes

- Inertia and the First Law
- Resists change in motion
- Needs an external unbalanced force
- Diagram: Block on a surface with force arrows
- Formula: \vec{F} = m\vec{a}

Activities

Coin and card (12 min, demonstration)

Materials: A stiff card, A coin, A glass

- Balance the card on the glass with the coin on top.
- Flick the card horizontally and have students predict what happens.

Student instructions:

- Predict, then observe, then explain using inertia.

Success looks like: Student predicts the coin falls into the glass.; Student explains it using resistance to a change in motion.

Support: Provide a sentence frame: 'The coin stayed still because...'

Extension: Ask what changes if the card is flicked slowly, and why.

Checkpoint Questions

- [understand] A book rests on a table. Is a force acting on it?

Expected: Yes — gravity and the normal force, which balance to zero net force.

Exit Ticket

In one sentence, why do passengers lurch forward when a bus brakes?

Success indicator: Answer refers to continued motion, not to a forward force.

Homework

- Find two examples of inertia at home and explain each in one sentence.

Estimated time: 20 min

Mentor Moment — Newton in a plague year

Newton developed much of his work on motion while Cambridge was closed during an outbreak, working alone at his family home.

Takeaway: Progress often comes from steady thinking in unpromising conditions.

Period 2

Entry Ticket

Name one situation where something kept moving after you stopped pushing it.

Expected response: Any example of continued motion, e.g. a rolling ball.

Teacher Script

0-5 min: Entry ticket

Collect two or three answers without correcting them yet.

Board: Write the three best student examples in a column.

5-20 min: Force, Mass and Acceleration

Introduce the idea using the student examples already on the board, then state the law formally.

Board: Write the formal statement beneath the examples.

- Anticipate: Why does a rolling ball eventually stop?

20-32 min: Activity

Run the activity; circulate and listen for the misconception.

32-37 min: Checkpoint

Pose the checkpoint question; take a show of hands before answers.

37-40 min: Exit ticket

Collect exit tickets at the door.

Blackboard Notes

- Force, Mass and Acceleration
- Resists change in motion
- Needs an external unbalanced force
- Diagram: Block on a surface with force arrows
- Formula: \vec{F} = m\vec{a}

Activities

Computing acceleration (12 min, problem_set)

Materials: Worksheet, Calculator

- Work the first item aloud, then release students to pairs.

Student instructions:

- Solve for a given F and m; state units at every step.

Success looks like: Correct value with correct units on at least three items.

Support: Provide the rearranged formula $a = F/m$ on the sheet.

Extension: Introduce a two-force problem requiring a net force first.

Checkpoint Questions

- [understand] A book rests on a table. Is a force acting on it?

  Expected: Yes — gravity and the normal force, which balance to zero net force.

Exit Ticket

In one sentence, why do passengers lurch forward when a bus brakes?

Success indicator: Answer refers to continued motion, not to a forward force.

Homework

- Find two examples of inertia at home and explain each in one sentence.

Estimated time: 20 min

Mentor Moment — Newton in a plague year

Newton developed much of his work on motion while Cambridge was closed during an outbreak, working alone at his family home.

Takeaway: Progress often comes from steady thinking in unpromising conditions.

Learning Gaps & Misconceptions

[high] Motion requires a continuously applied force.

- Diagnostic: A puck slides on frictionless ice. What force keeps it moving?

Reveals: Naming any force shows the misconception is still held.

- Remediation: Contrast a low-friction and a high-friction surface side by side.

Why: Isolates friction as the cause of stopping, not the absence of force.